



Lab Report

Khayyam AI Innovation Lab

Title

Breast Cancer Classification using miRNA Expression and Clinical Data (TCGA-BRCA)

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Abstract

This project evaluates whether miRNA expression profiles combined with clinical data from TCGA-BRCA can predict lymph node status (N0 vs N1). After preprocessing and scaling, we applied PCA for visualization and trained three classifiers: LDA, QDA and Gaussian Naive Bayes (GNB). Key evaluation metrics (accuracy, precision, recall, F1-score) were computed on a held-out test set.

1. Introduction

Breast cancer staging and early detection of metastasis is critical for treatment planning. miRNA levels have shown promise as biomarkers for tumor progression. This study aims to classify lymph node involvement using miRNA expression and basic clinical features from TCGA-BRCA.

2. Methods

Data sources: TCGA-BRCA miRNA expression file and clinical file (merged by sample ID).

Preprocessing:

- Remove duplicate samples.
- Truncate sample IDs to first 12 characters to align patients.
- Impute missing values using mean imputation.
- Standardize features with StandardScaler.
- Keep only samples with `pathology_N_stage == N0 or N1`.
- Train/test split: 80% train - 20% test.

Models trained:

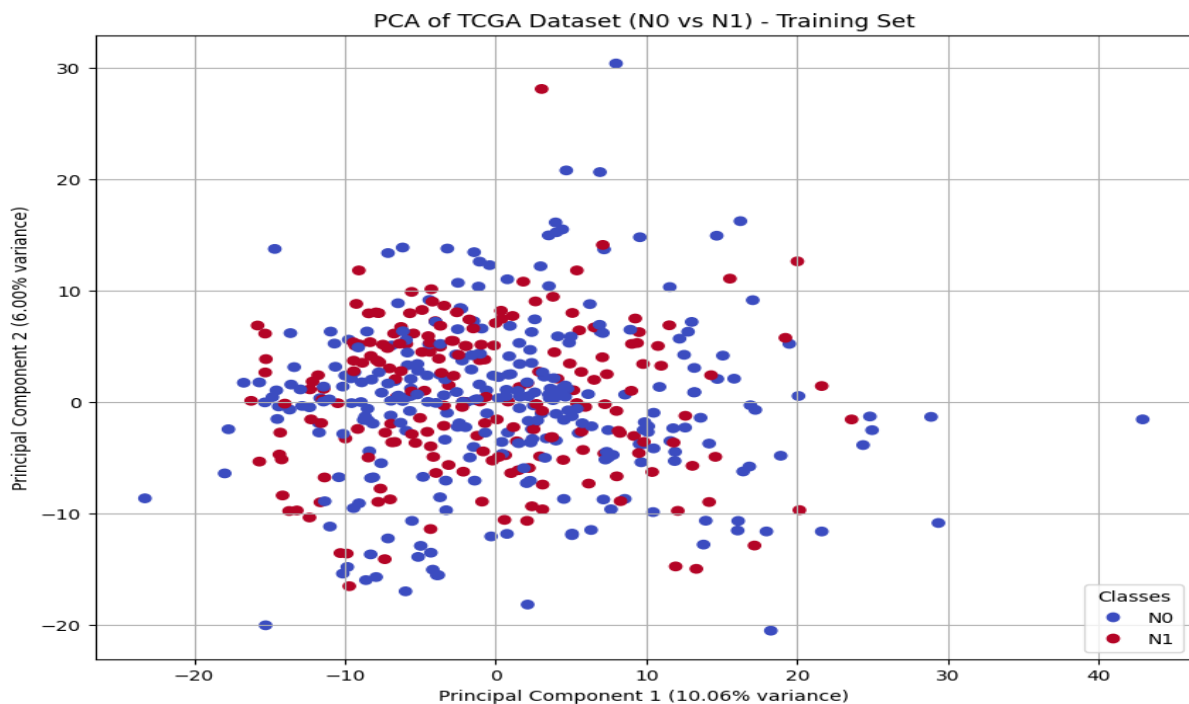
- PCA (visualization only)
- Linear Discriminant Analysis (LDA)
`sklearn.discriminant_analysis.LinearDiscriminantAnalysis`
- Quadratic Discriminant Analysis (QDA)
`sklearn.discriminant_analysis.QuadraticDiscriminantAnalysis`
- Gaussian Naive Bayes (GNB) `sklearn.naive_bayes.GaussianNB`

Evaluation metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix.

3. Results

3.1 PCA

- **Explained variance:** PC1 = 0.1006 , PC2 = 0.0600
- **Comment:** The first two principal components explain about 16 % of the total variance in the data. The PCA scatter plot shows that samples from classes N0 and N1 are partially mixed, indicating limited but visible separability between the two groups.
- **Figure:**



3.2 Linear Discriminant Analysis (LDA)

Training performance

- Accuracy (train): **0.94**

Test performance

- Accuracy (test): **0.56**

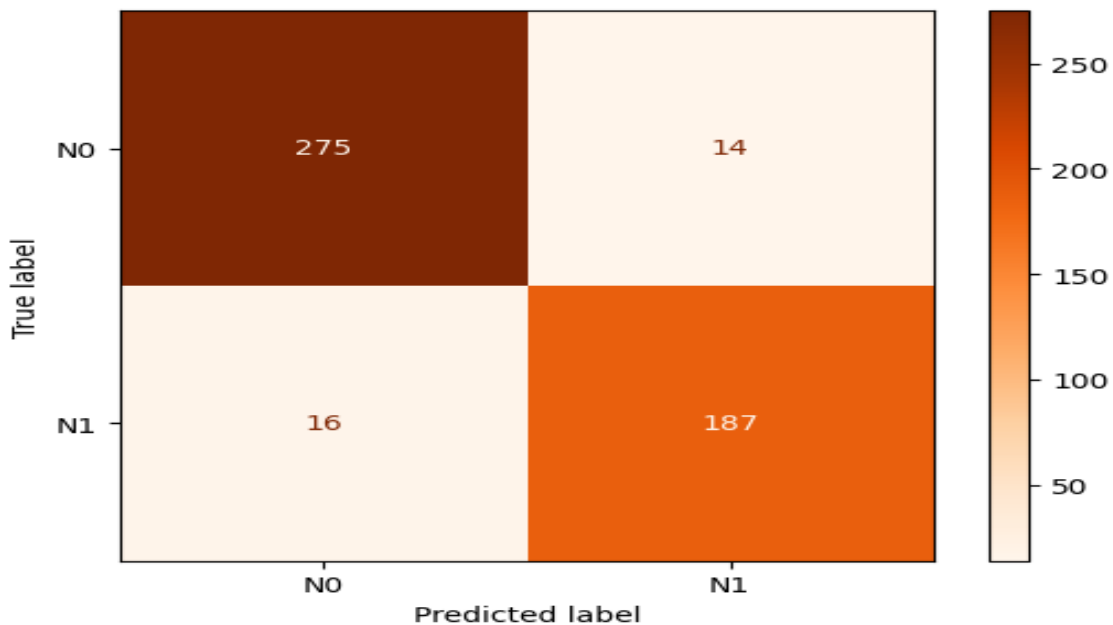
Classification report (test)

Class	Precision	Recall	F1-score	Support
N0	0.60	0.60	0.60	68
N1	0.51	0.51	0.51	55
Macro avg	0.56	0.56	0.56	123
accuracy			0.56	123

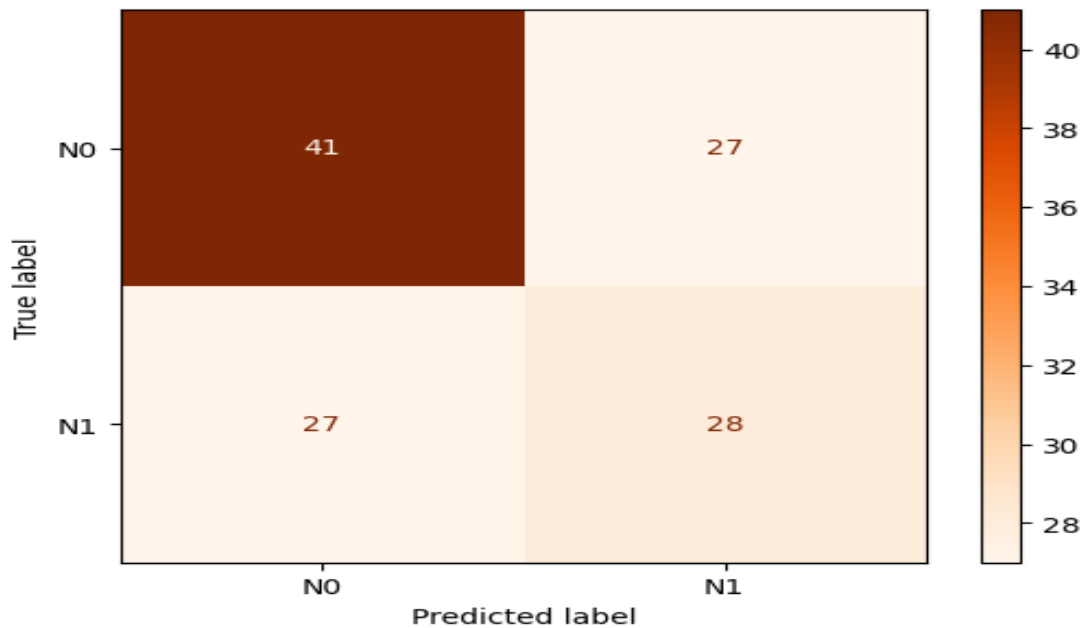
Classification report (train)

Class	Precision	Recall	F1-score	Support
N0	0.95	0.95	0.95	289
N1	0.93	0.92	0.93	203
Macro avg	0.94	0.94	0.94	492
accuracy			0.94	492

Confusion matrix (train)



Confusion matrix (test)



3.3 Quadratic Discriminant Analysis (QDA)

Training performance

- Accuracy (train): 1.0

Test performance

- Accuracy (test): 0.50406

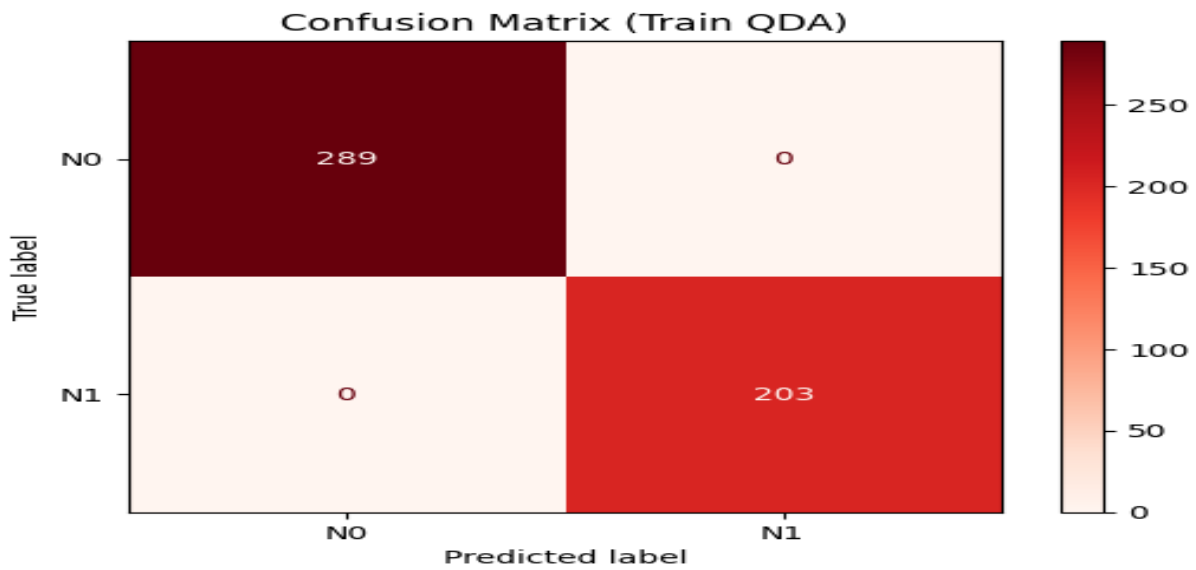
Classification report (test)

Class	Precision	Recall	F1-score	Support
N0	0.56	0.51	0.53	68
N1	0.45	0.49	0.47	55
Macro avg	0.51	0.50	0.51	123
accuracy			0.50	123

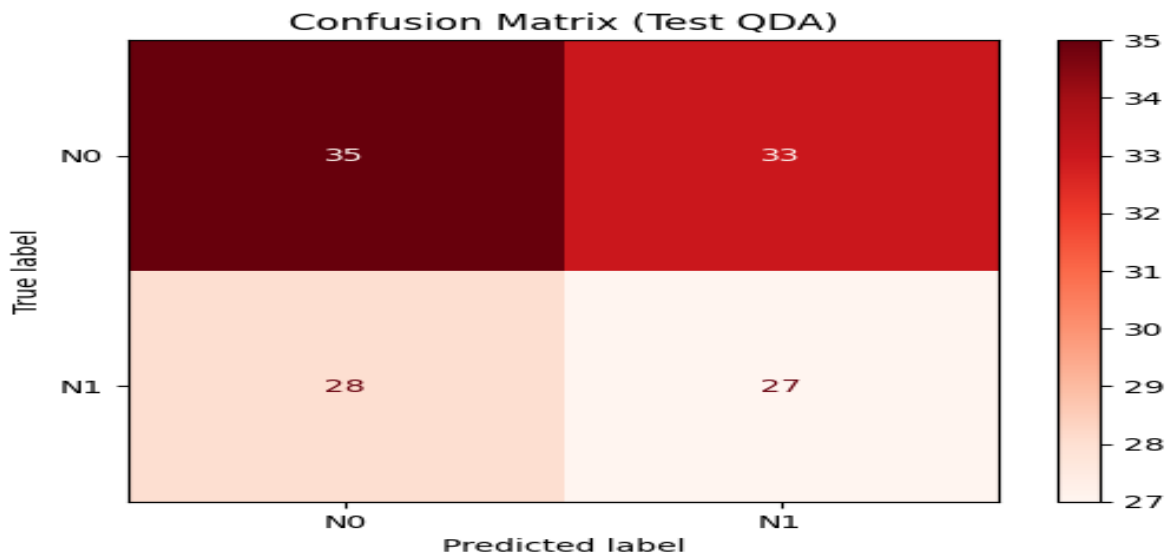
Classification report (train)

Class	Precision	Recall	F1-score	Support
N0	1.00	1.00	1.00	289
N1	1.00	1.00	1.00	203
Macro avg	1.00	1.00	1.00	492
accuracy			1.00	492

Confusion Matrix (train)



Confusion Matrix (test)



3.4 Gaussian Naive Bayes (GNB)

Training performance

- Accuracy (train): 0.58739

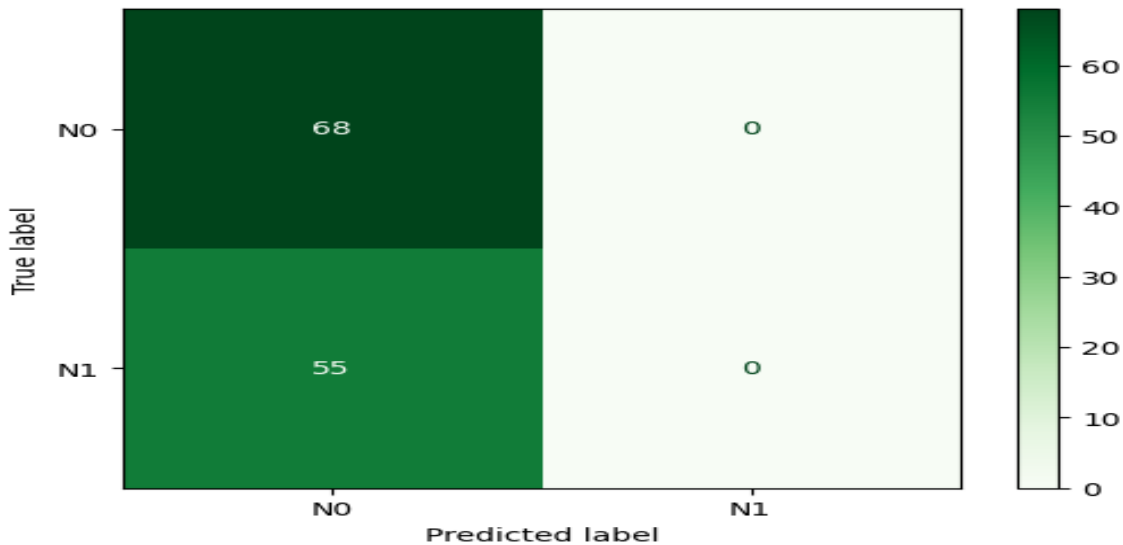
Test performance

- Accuracy (test): 0.55284

Classification report (test)

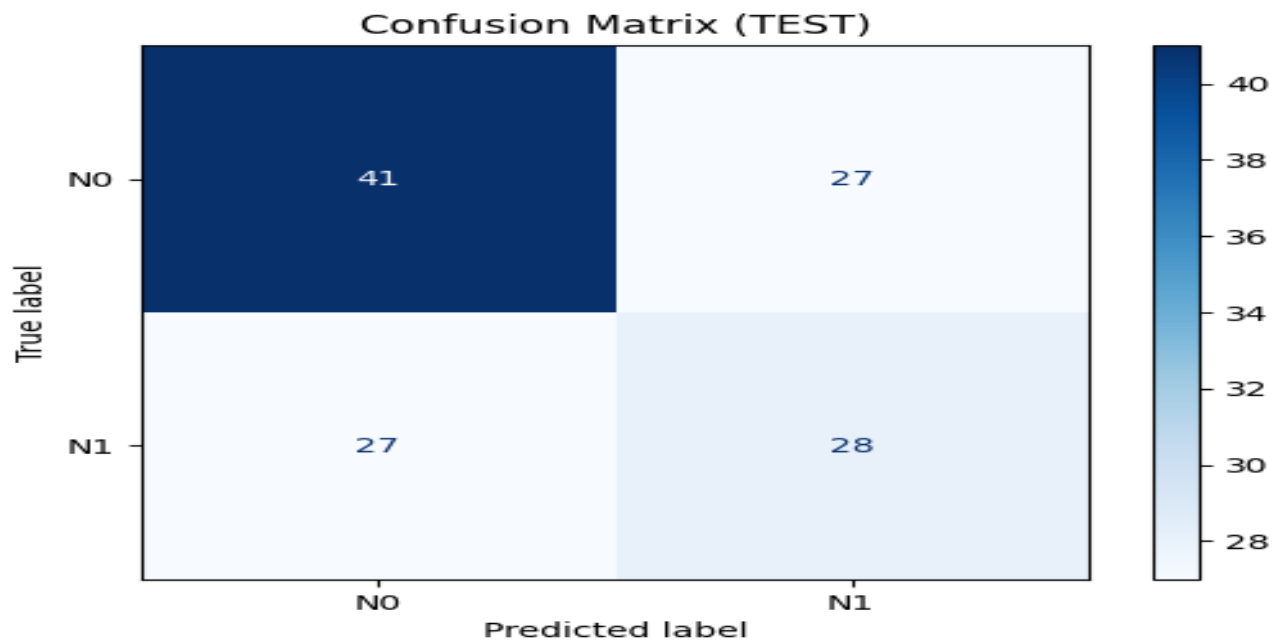
Class	Precision	Recall	F1-score	Support
N0	0.55	1.00	0.71	68
N1	0.00	0.00	0.00	55
Macro avg	0.28	0.50	0.36	123
accuracy			0.55	123

Confusion matrix (test) before PCA

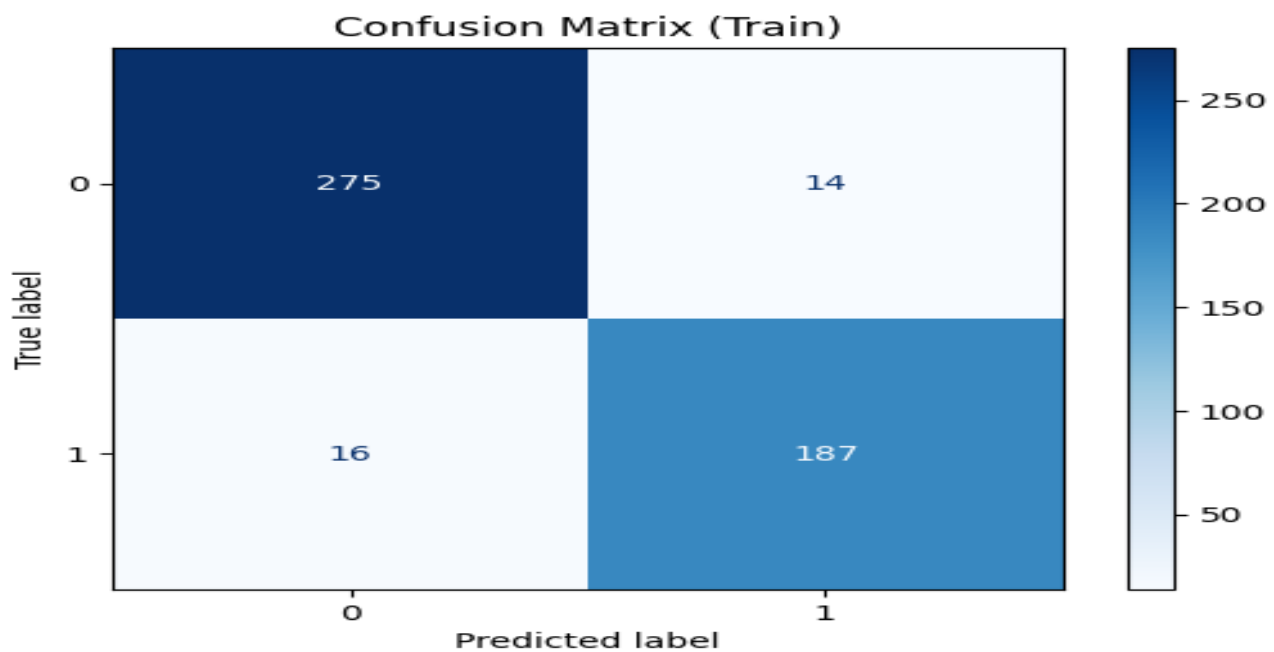


The confusion matrix shows that the model predicts all samples as class N0, indicating over-simplified decision boundaries when trained on high-dimensional data without PCA.

Confusion matrix (test) after PCA



Confusion matrix (train) after PCA



After applying PCA, the GNB model achieves more balanced predictions between N0 and N1 classes, improving recall for N1 and indicating that dimensionality reduction helped reduce bias toward class N0.

4. Discussion

Among the three classifiers, LDA achieved balanced performance between training and test accuracy, while QDA tended to overfit the training data. GNB showed lower accuracy but provided a fast baseline model. The PCA results indicated partial separation between the classes, suggesting that miRNA expression contains discriminative information related to lymph node status.

5. Conclusion

This study demonstrated that miRNA expression data combined with clinical features can predict lymph node status (N0 vs N1) with moderate accuracy. LDA provided the most stable results, while QDA achieved perfect training accuracy but lower generalization. Future work should focus on feature selection and external validation to improve robustness.